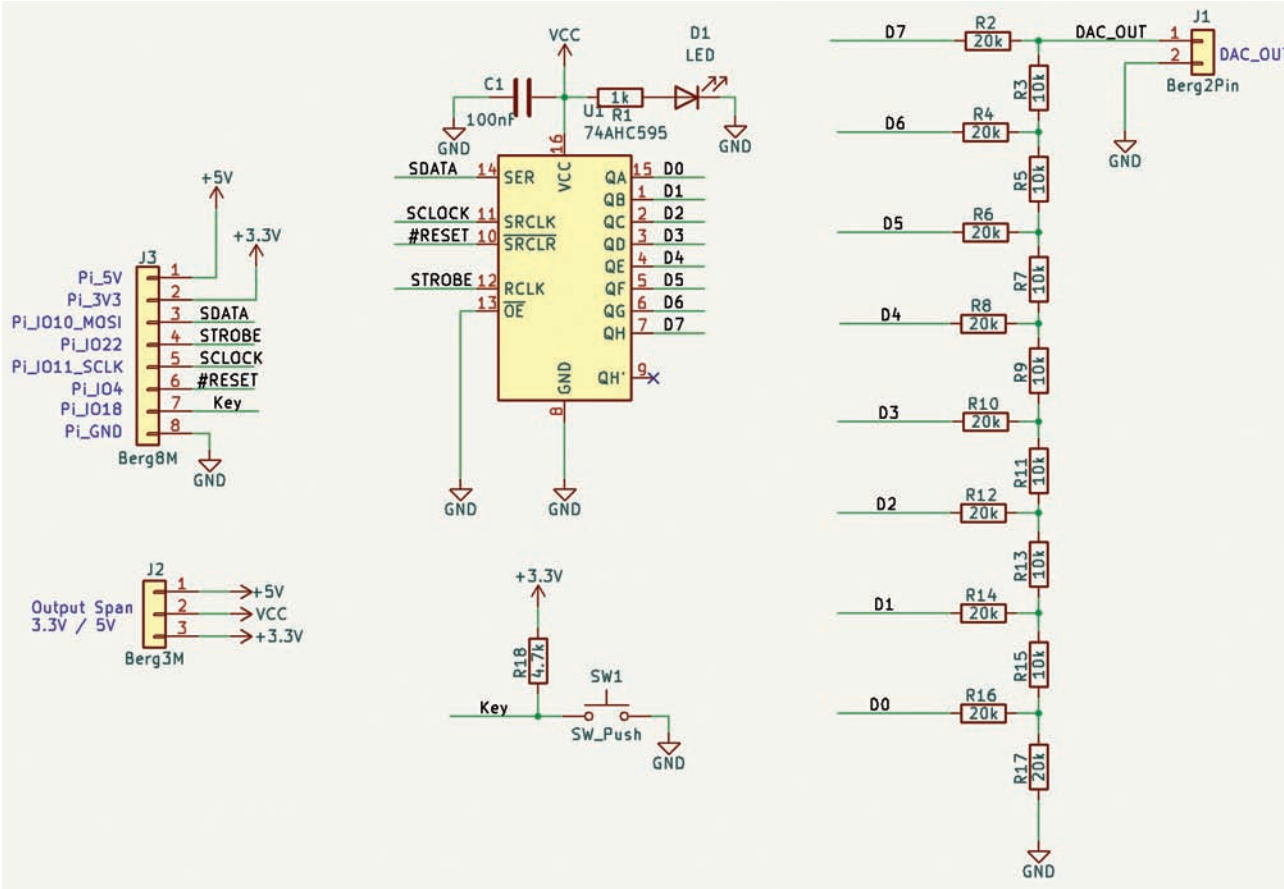

Fig. 2: Circuit diagram of the Raspberry Pi-based waveform generator

TABLE 2
RASPBERRY-PI GPIO PINS AND CONNECTIONS TO THE WAVEFORM GENERATOR DEVICE

Wave Generator Connections (J3)	Raspberry-Pi Function	Connector Pin No.		Pi-Function	Wave Generator Connections (J3)
J3 / Pin2	+3.3V	1	2	+5V	J3 / Pin1
	GPIO-2	3	4	+5V	
	GPIO-3	5	6	GND	J3 / Pin8
J3 / Pin6	GPIO-4	7	8	GPIO-14	
	GND	9	10	GPIO-15	
	GPIO-17	11	12	GPIO-18	J3 / Pin7
	GPIO-27	13	14	GND	
J3 / Pin4	GPIO-22	15	16	GPIO-23	
	+3.3V	17	18	GPIO-24	
J3 / Pin3	GPIO-10	19	20	GND	
	GPIO-9	21	22	GPIO-25	
J3 / Pin5	GPIO-11	23	24	GPIO-8	
	GND	25	26	GPIO-7	

Extended pins from 27 to 40 are not used, and the above connections are compatible with earlier versions of the Pi boards.

The coloured pins listed are actually used for the present waveform generator device.

Besides these two main serial port signals, only two other signals are needed to operate this DAC circuit. The 74HC595 shift register requires a 'strobe' signal that latches the serial data from the shift register to the output latches, presenting this data to the DAC input as 8-bit parallel data (D7-D0). This strobe signal is generated under software control using GPIO pin 22. There is also a global 'reset' signal that resets the output of the shift register to logic 0. This reset is applied only at the beginning and is controlled using GPIO pin 4.

Connector J2 is used as a switch to power the shift register with either 3.3V or 5V. This will switch the output data to 3.3V logic or 5V logic, thus allowing the full-scale output of the DAC (the 'output